LITERATURE SURVEY

Improving the Reliability of Quantum Circuits Using Digital Zero-Noise Extrapolation

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Project: Quantum Error Mitigation for Noisy Quantum Computing Systems

This literature survey is prepared specifically for the Digital Zero-Noise Extrapolation (Digital ZNE) project and follows the literature-survey content supplied in the project document. The first paper is the primary research paper supplied for the project; the remaining papers cover closely related quantum-error-mitigation directions.

1. Introduction

Quantum computing in the NISQ (Noisy Intermediate-Scale Quantum) era faces gate errors, decoherence, calibration uncertainty and hardware limitations. These noise sources reduce the accuracy and reliability of quantum computation results. Quantum Error Mitigation (QEM) provides practical techniques to improve computational results without requiring fully fault-tolerant quantum computers.

Digital Zero-Noise Extrapolation (ZNE) addresses this problem by executing a quantum circuit at different effective noise levels and extrapolating measured results toward the zero-noise limit. The project uses unitary folding to intentionally increase the noise level through logically cancelling operations. Linear, polynomial, Richardson, exponential and adaptive extrapolation methods are considered for comparison.

The proposed evaluation compares unmitigated and mitigated results using absolute error, expectation-value deviation, error reduction, gate count and number of circuit executions.

2. Papers Reviewed

2.1 Paper 1 — Giurgica-Tiron et al. (2021)

Digital zero noise extrapolation for quantum error mitigation

Category	Literature Summary
Problem Statement	NISQ processors produce noisy results, while scalable fault-tolerant error correction is not yet broadly available.
Quantum Approach / Algorithm	Digital ZNE, unitary folding, parameter noise scaling, and polynomial, linear, Richardson, exponential and adaptive extrapolation.
Key Findings / Results	The study reports strong error reduction in benchmark experiments and demonstrates ZNE on randomized circuits and a QAOA/MAXCUT example. It also shows that extrapolation performance varies by method.
Relevance to Our Project	This is the direct foundation for our project. It supplies the ZNE workflow, scaling ideas and comparison methods.

2.2 Paper 2 — Temme et al. (2017)

Error mitigation for short-depth quantum circuits

Category	Literature Summary
Problem Statement	Short quantum circuits can still accumulate errors that affect observable estimates.

Category	Literature Summary
Quantum Approach / Algorithm	Probabilistic error cancellation using quasi-probability based mitigation.
Key Findings / Results	Noisy quantum computations can be improved through classical post-processing, while the method introduces sampling overhead.
Relevance to Our Project	Provides a comparison point for ZNE: both approaches mitigate errors without requiring full fault-tolerant correction, but use different principles.

2.3 Paper 3 — Li et al. (2017)

Efficient variational quantum simulator incorporating active error minimization

Category	Literature Summary
Problem Statement	Variational quantum algorithms are sensitive to hardware errors and can produce degraded optimization results.
Quantum Approach / Algorithm	Error-mitigation ideas applied to variational quantum computation.
Key Findings / Results	The work demonstrates that error-aware processing can improve the usefulness of variational computations on noisy hardware.
Relevance to Our Project	Supports testing ZNE with variational circuits rather than only simple benchmark circuits.

2.4 Paper 4 — Cai et al. (2023)

Quantum error mitigation

Category	Literature Summary
Problem Statement	Different QEM methods have different assumptions, resource costs and applicability to noise sources.
Quantum Approach / Algorithm	Review of ZNE, probabilistic error cancellation, symmetry-based methods, virtual distillation and related QEM techniques.
Key Findings / Results	Provides a broad framework for understanding the capabilities and limitations of modern quantum error mitigation.
Relevance to Our Project	Provides the wider literature foundation for positioning our ZNE comparison among modern QEM techniques.

3. Comparison of the Reviewed Work

Paper	Focus	Type of Evidence	Main Limitation / Observation
Giurgica-Tiron et al.	Digital ZNE and noise scaling	Benchmark experiments	Directly relevant; performance varies across extrapolation methods.
Temme et al.	Probabilistic error cancellation	Error-mitigation methodology	Introduces sampling overhead and uses a different mitigation principle.
Li et al.	Error mitigation for variational computation	Variational quantum computation	Supports evaluation on variational circuits.
Cai et al.	Broad quantum error mitigation	Review / synthesis	Shows that QEM methods have different assumptions and resource costs.

4. Research Gap and Project Motivation

The reviewed literature shows that quantum error mitigation can improve noisy quantum computations, but the effectiveness of a mitigation method depends on the circuit, noise model, scaling factors and measurement budget. No single extrapolation method is guaranteed to dominate in every situation.

This motivates a comparative implementation in which multiple circuit types and multiple ZNE extrapolation methods are evaluated using common accuracy and resource metrics. The project uses Digital ZNE as a software-level mitigation layer: unitary folding increases effective noise while preserving the logical operation, and classical extrapolation estimates the zero-noise result.

5. Limitations Identified in the Literature

• Noise scaling increases the number of circuit operations or executions, introducing computational and measurement overhead.

• Extrapolation is an estimation problem; statistical fluctuations can make high-order models unstable.

• Unitary folding is not equally suitable for every error source, especially state-preparation and measurement errors.

• Coherent or systematic errors may not be amplified in the same way as incoherent noise.

• The best extrapolation method depends on the circuit, noise model, scaling factors and measurement budget.

6. Conclusion

The literature establishes Digital Zero-Noise Extrapolation as a practical quantum error mitigation approach for NISQ-era computations. Digital ZNE uses controlled noise scaling and classical inference to estimate a less noisy result. The primary research paper demonstrates unitary folding and different extrapolation strategies, while the related literature places ZNE within the broader QEM field. The proposed project focuses on measuring practical error reduction and identifying the trade-off between improved accuracy and additional resources.

7. References

[1] T. Giurgica-Tiron, Y. Hindy, R. LaRose, A. Mari and W. J. Zeng, "Digital zero noise extrapolation for quantum error mitigation," arXiv:2005.10921.

[2] K. Temme et al., "Error mitigation for short-depth quantum circuits," 2017.

[3] Y. Li et al., "Efficient variational quantum simulator incorporating active error minimization," 2017.

[4] Z. Cai et al., "Quantum error mitigation," 2023.